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TMA4268 Cheat Sheet

Bias–variance & double descent

$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \underbrace{(f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2}_{\text{Bias}^2} + \text{Var}(\hat{f}(x_0)) + \sigma^2.$

Outer $\mathbb{E}$: over training set *and* test noise ε_0. Inner Bias/Var: training set only. Cross-terms vanish: (i) $\varepsilon_0 \perp \hat{f}$, $\mathbb{E}\varepsilon = 0$; (ii) $\mathbb{E}[\hat{f} - \mathbb{E}\hat{f}] = 0$. Identity is *exact* for any estimator. **[double descent]**: in $p \gg n$, SGD/pseudoinverse picks min-$\|\theta\|_2$ interpolator $\Rightarrow$ implicit ridge $\Rightarrow$ variance *can fall* past interpolation peak. Bias+Var sum still holds. Prof's rename: "avoid bad overfitting" (not trade-off): regularization cuts var without paying full bias.

Shrinkage MSE. For $\hat{\mu}_c = c\bar{y}$: $c^\star = \mu^2/(\mu^2 + \sigma^2/n)$.

Linear regression

$\hat{\beta} = (X^\top X)^{-1} X^\top y$. $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$. $\text{df}_{\text{model}} =$ #params incl. intercept; K-level factor $\to$ $K - 1$ dummies. **CI vs PI.** CI: $\hat{y}_0 \pm t\hat{\sigma}\sqrt{x_0^\top (X^\top X)^{-1} x_0}$; PI: $\hat{y}_0 \pm t\hat{\sigma}\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}$ ($+1$ = irreducible). ≈ 1.96 for large $n - p - 1$. **R^2 vs adj.** $R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1}$. R^2 monotone in p; adj may drop; large gap $\Rightarrow$ over-parameterized. **MLE $\equiv$ LS.** under $\varepsilon \sim N(0, \sigma^2)$: $\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(y_i - x_i^\top \beta)^2$, drop constants $\Rightarrow$ $\arg\max = \arg\min$ RSS. Laplace errors $\Rightarrow$ L1 (LAD), robust. **Diagnostics.** resid vs fit (linearity/var); QQ (normality **[prof favors]**); scale-loc (heterosc.); resid vs leverage. Fanning $\Rightarrow$ log-y or WLS.

Collinearity

Perfect: $X^\top X$ singular $\Rightarrow$ OLS *undefined* (**[not "infinite SE"]**); software drops a column. **Near**: $(X^\top X)^{-1}$ inflates diagonal in collinear direction; SE explode but $\hat{y} = X\hat{\beta}$ stable. Pairs have large negative $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k)$ that cancels in joint contribution. **Fingerprint.** individually insig. t + significant joint F + drop-one $\approx$ same test MSE. **[Standardization does NOT fix collinearity]** (rank unchanged); only helps shrinkage scale-fairness. Ridge fix: $(X^\top X + \lambda I)^{-1}$ exists $\forall\lambda > 0$; splits coef across collinear pair. Lasso: corner $\Rightarrow$ picks one. VIF $\approx 1/(1 - R_j^2)$.

Interactions & categorical

With $x{:}z$ in model: β_z = effect of z *at* $x = 0$ only (**[often outside data]**; centre x for sane interpretation). Slope of x at level z: $\beta_x + \beta_{xz} z$. **Hierarchy principle**: keep main effects whenever interaction stays in. Re-encoding/reference flip: RSS, fits, $\hat{p}$ *invariant*; coefficients and SEs change. **[Dummy trap]**: K dummies + intercept $\Rightarrow$ singular (sum to 1). Use $K - 1$ dummies. Drop one; ridge does *not* fix this structurally.

Logistic regression

$\eta = \beta^\top x$, $\hat{p} = \sigma(\eta) = \frac{e^\eta}{1+e^\eta}$. Odds $= p/(1 - p)$; $p = O/(1 + O)$. $\exp(\beta_j \Delta)$ = odds-ratio per Δx_j. **[β_j is on log-odds, NOT probability]**: $\partial p/\partial x_j = \beta_j p(1 - p)$; max slope $\approx 0.25\beta$ at $p = 0.5$. $p_{\text{new}} = \frac{OR \cdot p}{1-(1-OR)p}$ (only $\approx OR \cdot p$ for tiny p). **Interaction** $x{:}z$. odds factor per Δx=1 at level z: $\exp(\beta_x + \beta_{xz} z)$ (per-group multipliers; sign can flip with z). Predicted p under group switch: add *both* dummy and interaction terms.

LDA / QDA / Bayes

$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log\pi_k$ (LDA: shared Σ, drops $-\frac{1}{2}x^\top\Sigma^{-1}x$ as k-free $\Rightarrow$ linear boundary). QDA: $\delta_k = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1}(x - \mu_k) + \log\pi_k$ (per-class Σ_k $\Rightarrow$ quadratic boundary). **2D boundary recipe.** $f_A\pi_A = f_B\pi_B$: $v = \Sigma^{-1}(\mu_A - \mu_B)$; $c = -\frac{1}{2}\mu_A^\top\Sigma^{-1}\mu_A + \frac{1}{2}\mu_B^\top\Sigma^{-1}\mu_B + \log(\pi_A/\pi_B)$; line $v^\top x + c = 0$. Sanity: equal priors $\Rightarrow$ midpoint on line. **Prior shift.** $\pi_k \uparrow$ moves boundary *away* from class k (its decision region *grows*); equivalently, more area labelled k. Boundary shift along v: $\frac{1}{\|v\|^2}\log(\pi_A/\pi_B)$. **Small n**: LDA > QDA (per-class Σ_k has $Kp(p + 1)/2$ params, noisy). Binary features violate Gaussian assumption $\Rightarrow$ logistic safer. **Naive Bayes** $\equiv$ diagonal-Σ LDA. **[Mixture class density]**: $f_0 = \sum w_m \phi(\cdot; \mu_m, \Sigma)$ $\Rightarrow$ boundary neither linear nor quadratic (log of sum-of-exps doesn't collapse) – LDA/QDA fail.

Resampling: CV, bootstrap

k-fold $\text{df}_k = \frac{1}{k}\sum_k \text{MSE}_k$, $\widehat{\text{SE}} = \text{sd}(\text{MSE}_k)/\sqrt{k}$ (**["not quite valid"]**: folds share data). **LOOCV**: **[low bias, HIGH variance]** (overlapping train sets correlate fold errors). k=5, 10 usually wins on variance. **OLS LOOCV shortcut.** $\text{CV}_n = \frac{1}{n}\sum(\frac{y_i - \hat{y}_i}{1-h_{ii}})^2$, $H = X(X^\top X)^{-1}X^\top$. OLS-only. **1-SE rule.** $\lambda_{1\text{SE}} = \max\{\lambda : \text{CV}(\lambda) \le \text{CV}(\hat{\lambda}_{\min}) + \widehat{\text{SE}}\}$; pick *simplest* (**[$\lambda$ ↑, deg↓, k_{KNN} ↑]**). **Wrong-way CV.** anything using y (feature screen, supervised PCA) *must* be *inside* each fold. Else CV biased ↓ (toward 0% on noise w/ $p \gg n$). **Nested CV.** inner = tuning, outer = honest assessment. Naive $\min_\theta \text{CV}_k$ is winner's-curse biased ↓. Random k-fold **breaks** on time series $\Rightarrow$ rolling-origin / blocked. Validation set $\neq$ 2-fold. **Bootstrap.** sample n *with replacement*; $\widehat{\text{SE}} = \text{sd}(\hat{\theta}^{*b})$. $B \in [10^3, 10^4]$. $P(i \in \text{boot}) = 1 - (1 - 1/n)^n \to 1 - 1/e \approx 0.632$; OOB $\to 1/e \approx 0.368$. Percentile 95% CI: $[\hat{\theta}^*_{(.025)}, \hat{\theta}^*_{(.975)}]$ (order stats, **[not $\hat{\theta} \pm 1.96$SE]**). Quantifies *variance*, *not* bias. Paired (case) bootstrap for random-X/observational; residual bootstrap needs correct mean and homoscedastic resids. Boot-SE $\neq$ GLM-SE $\Rightarrow$ GLM assumptions violated, *not* a bootstrap flaw.

Ridge / Lasso / PCR

Ridge $\min \|y - X\beta\|^2 + \lambda\sum_{j\ge1}\beta_j^2$ (intercept *unpenalised*); closed form $(X^\top X + \lambda I)^{-1}X^\top y$ on centered X. Lasso $\min \frac{1}{2n}\|\cdots\|^2 + \lambda\|\beta\|_1$; corners on axes $\Rightarrow$ exact zeros. **Geometry.** L^2 ball smooth $\Rightarrow$ splits collinear pair; L^1 diamond $\Rightarrow$ corner picks one. **[Lasso path depends on dummy reference]**; use group lasso for factors. **Std before:.** ridge, lasso, PCA, PCR, KNN, k-means, hier. clust. OLS doesn't need it. **Subset counts.** best subset 2^p; forward (full path) $1 + p(p + 1)/2$; forward to size K: $1 + \sum_{j=0}^{K-1}(p - j)$. Backward fails in $p > n$. **PCR = discretised ridge.** SVD: ridge shrinks PC-j by $d_j^2/(d_j^2 + \lambda)$ (smooth); PCR keeps top M at weight 1, rest 0. PCR not variable selection. Lasso > PCR for sparsity in original predictors. **[Lasso $\neq$ AIC step]**: avoids *model-selection bias*; AIC+refit gives anti-conservative p-values; BIC does not fix it. $p > n$ feasibility: ridge ✓, lasso ✓ (selects $\le n$), forward ✓, backward **[×]**, boosting ✓.

Splines / GAM

df counts (excl. global intercept): polynomial d: d; step K-cut: K; linear spline: $K + 1$; **cubic spline:** $K + 3$; **natural cubic:** $K + 1$; smoothing spline: $\text{tr}(S_\lambda) \in [2, n]$, non-integer. Cubic spline is C^2 at knots; **[3rd derivative discontinuous]**; knots fixed in advance. **Smoothing spline penalty.** $\int g''(t)^2 dt$ – on *second* derivative, **[not first]**. $\lambda \uparrow\Rightarrow$ smoother, df ↓; $\lambda \to \infty \Rightarrow$ df = 2 (line). Opposite direction from poly degree / knot count. **GAM total df.** $1 + \sum_j \text{df}_j + \sum_c (L_c - 1)$ (cat. levels). **GAM limitation.** $\sum f_j(x_j)$ has *no interactions* unless tensor-product added; trees of depth $d \ge 2$ get them free. Boosting $\gg$ GAM gap $\Rightarrow$ interactions present.

PCA

Loadings ϕ_m unit-norm; score $z_{im} = \phi_m^\top x_i$ (no recentering if x already standardised). Total var $= \sum\lambda_j = p$ on std. data; $\text{PVE}_m = \lambda_m/p$. PC1 *maximises* variance; sign arbitrary. **[NOT scale-invariant]** – largest-variance variable dominates PC1 if unscaled. **[PCA blind to y]** (PLS is the supervised cousin).

Trees / Bagging / RF

$\text{Var}(\bar{f}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ **[(prof-derived, not in ISLP)]**; floor $\rho\sigma^2$ as $B \to \infty$. **RF** decorrelates: `mtry` $< p$ shrinks ρ. Defaults: $\sqrt{p}$ classif, $p/3$ regr. B **[NOT a tuning parameter]** in RF: test err monotone ↓ in B; pick $\approx$ 500-1000. Trees grown deep, no pruning. Bagging = RF with `mtry` $= p$. **OOB.** each tree excludes $\approx 1/e \approx 37\%$; predict i from only trees that didn't see i $\Rightarrow$ near-unbiased test-err estimate, free. **Variable importance.** magnitude only **[(no direction, shape, causality)]**; pair with PDP/ICE. Prof prefers **permutation** (drop in OOB acc after shuffle) over Gini-impurity (biased toward high-cardinality). **Regression tree leaf prediction.**: mean of training y_i in region.

Boosting (AdaBoost / GB / XGBoost)

AdaBoost. weight $w_i^{(m+1)} \propto w_i^{(m)}\exp(\alpha_m \mathbf{1}[y_i \neq G_m])$, renormalise. $\alpha_m = \log(\frac{1-\text{err}_m}{\text{err}_m})$ (or $\frac{1}{2}$ log; consistent w/ slide). err$= 0.5 \Rightarrow \alpha = 0$; err$> 0.5 \Rightarrow \alpha < 0$ **[(vote auto-flips)]**. **Derivation (prof-flagged)**: exp loss $L = e^{-yf}$, $y \in \{-1, +1\}$. Split sum into correct/wrong: $J(\alpha) = W_C e^{-\alpha} + W_W e^{\alpha}$; $dJ/d\alpha = 0 \Rightarrow e^{2\alpha} = W_C/W_W = (1 - \text{err})/\text{err}$. AdaBoost $\equiv$ forward-stagewise additive on exp loss. **Gradient boosting.** init $\hat{f}^{(0)} = \bar{y}$; fit tree to neg-grad $r_i = -\partial L/\partial f = y - f^{(m-1)}$ (squared loss $\Rightarrow$ residuals); update $\hat{f}^{(m)} = \hat{f}^{(m-1)} + \nu\hat{T}_m$. **Hyperparams**: M (*IS* tuned, **[overfits]** unlike RF); $d \in \{1 - 8\}$ (interaction order; $d = 1$ = additive); $\nu \in [10^{-3}, 10^{-1}]$. **Coupling**: $M \cdot \nu \approx$ const; halving ν doubles required M. **Bagging vs boosting.** bagging/RF reduce *variance*; boosting reduces *bias* (sequential). RF averages independent bootstraps; boosting sums correlated weak learners. **XGBoost.** $\Omega(T) = \gamma|T| + \frac{1}{2}\lambda\|w\|^2 + \alpha\|w\|_1$; gradient *and* Hessian (Newton/2nd-order); α drives small leaves to 0; γ rejects splits unless gain $> \gamma$; row+col subsampling.

Neural networks

Param count. per layer $(n_{\text{in}} + 1) \cdot n_{\text{out}}$ (*bias on receiving layer; inputs no bias*). E.g. $4 \to 5 \to 3 \to 1$ = 25+18+4 = 47. Dropout / early stop / label smooth add 0 params. Need non-linear activation (else $W_L \cdots W_1$ collapses to single affine $\Rightarrow$ linear regression). **ReLU.** $\max(0, z)$; $g'(z) = \mathbf{1}[z > 0]$ **[(dying ReLU)]**. Sigmoid: $\sigma' = \sigma(1 - \sigma)$. **Backprop seed.** $\delta^{\text{out}} = -(y - \hat{f})$ (squared loss). **Sigmoid + BCE** $\partial L/\partial z = \hat{p} - y$ (the $\hat{p}(1 - \hat{p})$ factors cancel $\Rightarrow$ no vanishing at saturation). Softmax + CE: $\delta_c^{\text{out}} = \hat{p}_c - y_c$. Chain for 1 hidden: $\partial L/\partial\alpha_{kj} = (\hat{p} - y)\beta_k g'(z_k)x_j$. Backprop = *gradient computation*; SGD/Adam does *update*. Needs DAG (RNN $\to$ BPTT). **Mini-batch SGD.** $\mathbb{E}[\widehat{\nabla L}] = \nabla L$ (unbiased); Var $\propto 1/m$. Bigger batch $\Rightarrow$ less noise $\Rightarrow$ *weaker* implicit reg. Pow-2 batch is hardware, not statistics. Implicit min-$\|\theta\|_2$ interpolator $\Rightarrow$ benign overfitting / flat minima (**[prof hobbyhorse]**). **Learning rate.** $\eta \in [10^{-3}, 10^{-1}]$ typical; $\eta = 0.1$ fine; $\eta \approx 2$ diverges. **Dropout.** train-only; rate $p \in [0.2, 0.5]$, default 0.2; **[NEVER 50% on tabular]** (prof's iron rule); test = full net w/ rescaling. **Label smoothing.** target $\tilde{y}_c = 1 - \varepsilon$ true, $\varepsilon/(C - 1)$ else; $\varepsilon \approx 0.05$. Motivation: label noise; discourages over-confident logits. **Early stopping.** on *validation* loss (**[not training]**); return params at val-min epoch. **L2 weight decay.** raises training err, lowers test err. "Always regularise NN" (SGD noise counts).

Classifier metrics / ROC

Sens = TP/P = TPR; Spec = TN/N = TNR; FPR = 1−Spec; Err = (FP+FN)/n; Prec = TP/(TP+FP). **ROC.** y-axis TPR, x-axis FPR = 1−Spec; diag = chance. Threshold ↓$\Rightarrow$ more positives flagged $\Rightarrow$ Sens↑, Spec↓, point moves **NE**. AUC $= P(\hat{s}(X^+) > \hat{s}(X^-))$; threshold-free. AUC = 0.5 random; < 0.5: flip predictions. **Imbalance.** naive baseline acc $= \max\pi_k$ ($\Rightarrow$ accuracy lies). Use (Sens, Spec), AUC, PR-AUC, F1. Asymmetric cost (medical screening): optimise *sensitivity*, lower threshold (**[0.5 inappropriate at base rate]** 0.2/0.8). **KNN.** std features (Euclid dominated by big-unit feat.); large K $\Rightarrow$ high bias.

Clustering

Hierarchical. complete $D(A, B) = \max d$; single = min; average = mean. **First merge is linkage-independent** (singletons). Recipe: find min off-diag $\to$ merge $\to$ recompute by rule. Heights non-decreasing. Cut at h: #vertical lines crossed = #clusters. Leaf horizontal order arbitrary. **[−1P per labelled mistake]** (incl. missing y-axis). Complete fuses $\ge$ single height for same pair. **K-means.** non-increasing WCSS, local min (random init $\to$ many starts). NOT nested across K. Std features. User chooses K.

Misc / direction-of-effect

[F-test]: joint test on $K - 1$ dummies for a K-level factor (use `anova`; not individual t's). Prof: *name* it, don't compute mechanics. Individual t's can all be insig. while joint F rejects (collinearity). **[Causal vs correlation]**: prof slogan "fancy correlations, not causal" – observational $\hat{\beta}$ are conditional associations, no matter how tiny the p. **[Validation set $\neq$ 2-fold]** (fits once vs twice). **[Practical $\neq$ statistical]** significance. Reducible (var of $\hat{f}$, can be cut by n) vs irreducible σ^2 (cannot). **Plot diagnostics.** (1) resid vs fit: zero-mean + homoscedasticity; (2) QQ: normality (**[prof favorite]**); (3) scale-location: het. in std-resid form; (4) resid vs leverage: influence (high lev + high resid = danger).

Worked numerics

NN params. $4\to5\to3\to1$: $(5)\cdot5+(6)\cdot3+(4)\cdot1 = 25+18+4 = 47$. $128\to32\to64\to5$: $129\cdot32 + 33\cdot64 + 65\cdot5 = 6533$. Forgetting biases = canonical wrong.
AdaBoost α. err$= 0.20 \Rightarrow \alpha = \log 4 \approx 1.386$. err$= 0.40 \Rightarrow \alpha \approx 0.405$. err$= 0.70 \Rightarrow \alpha \approx -0.847$ (flip). Weight mult: correct $\times e^{-\alpha}$, wrong $\times e^{+\alpha}$, renormalise.
OOB. n=5: $(4/5)^5$=0.328. n=1000: $\approx 1/e$, OOB count ≈ 368.
Sigmoid. $\sigma(0.4) = 0.599$; $\sigma(-1) = 0.269$. Odds$= 2.5 \to p = 5/7 \approx 0.714$. $p = 0.3 \to$ odds$= 3/7$. $\exp(0.8) \approx 2.225$.
Quadratic vertex. $\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$: minimum/max at $x^\star = -\beta_1/(2\beta_2)$ (min if $\beta_2 > 0$).
CI shortcut. large df: $\hat{\beta} \pm 1.96\,\widehat{\text{SE}}$.

Tuning / decision tables

Standardise before?. **Yes**: ridge, lasso, PCA/PCR, KNN, k-means, hier. clust., SVM. **No**: OLS, plain logistic, trees/RF/boosting.
$p > n$ feasibility. ridge ✓, lasso ✓ ($\leq n$ selected), forward step ✓, backward **[×]**, OLS **[×]**, GBM/RF/NN ✓.
Tuned by CV?. Ridge λ ✓; Lasso λ ✓; PCR M ✓; PLS M ✓; Smoothing-spline λ/df ✓; RF `mtry` ✓ (but B/ntree NO); GBM M, ν, d all ✓.
Method ranking gap. OLS$>$Ridge $\Rightarrow$ OLS variance-dominated; Ridge$<$GAM $\Rightarrow$ nonlinearities; GAM$<$Boost $\Rightarrow$ interactions present.
Sparsity. Lasso ✓; Ridge ×; PCR × (all original vars in every PC); Trees/Lasso/Best subset: yes.

Spline / GAM df table

Object	incl. int.	excl. int.
poly deg d	$d+1$	d
step, K cuts	$K+1$	K
linear spline K knots	$K+2$	$K+1$
cubic spline K knots	$K+4$	$K+3$
natural cubic K knots	$K+2$	$K+1$
smoothing spline	$\text{tr}(S_\lambda)$	—

K knots: natural saves 2 df via boundary linearity. `bs(x,df=)`: intercept absorbed in arg.

Plot reading / interpretation

ROC curve. 45° diag = chance. Knee toward NW = better. AUC = area; two curves can have equal AUC and different shapes. **PCA biplot.** arrows = loading vectors (ϕ_m); long y-arrow $\Rightarrow$ high PC2 loading. Point on axis means: that observation's score on that PC. PC1 dominated by variable with largest $|\phi_{j1}|$. **Scree / cumPVE.** elbow rule: keep PCs until cum. PVE plateaus, or to a target (80/90/95%). **Coefficient table.** sanity: $n-p-1$ = residual df. CI $\hat{\beta} \pm 1.96\widehat{\text{SE}}$. Insig. t + huge SE on a predictor with significant partner $\Rightarrow$ *collinearity*, not "no effect". **Lasso path.** coefs hit zero in order of weakness; reading sequence of zeros gives effective variable ranking. **CV plot.** u-shape with min at λ_{min}; band gives λ_{1SE} (largest λ within band = simplest).

Special cases / niche traps

Mixture-class density. class-0 $= \sum_m w_m \phi(\cdot; \mu_m, \Sigma)$. $\log(f_0 \pi_0)$ contains $\log \sum e^{\cdots}$ which *doesn't* collapse $\Rightarrow$ Bayes boundary neither linear nor quadratic; LDA *and* QDA both inadequate. **SVM "linear in features".** only if every feature in the boundary appears in the listed basis. Cubic terms (X_2^3) break "linear in $\{X_1, X_2, X_1^2, X_2^2\}$". To classify a point: plug in, check sign. **KNN K.** $K = 1$: zero train err, max variance. $K = n$: constant prediction, max bias. Std features. **K-means.** not nested across K (**[going $K \to K+1$ can reshuffle globally]**; only hier. is nested). **Sigmoid+SE vs Sigmoid+BCE.** BCE cancels $p(1-p)$ in gradient $\Rightarrow$ no vanishing at saturation. Use BCE. **Boosting init.** squared loss: $\hat{f}^{(0)} = \bar{y}$. Classification AdaBoost: weights $w_i^{(0)} = 1/n$. **Linkage height ineq..** for same pair: $h_{\text{complete}} \geq h_{\text{average}} \geq h_{\text{single}}$.

Direction-of-effect quick ref

$\lambda_{\text{ridge}} \uparrow$	$\hat{\beta} \to 0$, var$\downarrow$, bias$\uparrow$, flex$\downarrow$
$\lambda_{\text{smooth}} \uparrow$	smoother, df$\downarrow$; $\to \infty$ gives line
poly deg $\uparrow$	wigglier, df$\uparrow$, flex$\uparrow$
K knots $\uparrow$	wigglier, df$\uparrow$
K in KNN $\uparrow$	smoother, bias$\uparrow$, var$\downarrow$
`mtry` $\downarrow$ in RF	$\rho \downarrow$, var$\downarrow$
$\nu \downarrow$ in boost	need $M \uparrow$
$M \uparrow$ in boost	may overfit (**[not in RF]**)
batch size $\uparrow$ in SGD	noise$\downarrow$, implicit reg$\downarrow$
dropout $p \uparrow$	more reg, bias$\uparrow$ if too large
weight decay $\uparrow$	train err$\uparrow$, test err often$\downarrow$
threshold $\downarrow$	sens$\uparrow$, spec$\downarrow$, NE on ROC
$\pi_k \uparrow$ in LDA	boundary moves *away* from k
add predictor (signal)	test MSE may $\downarrow$
add predictor (noise)	test MSE often $\uparrow$
re-encode reference	fits invariant; $\hat{\beta}$ change

Per-question opening moves

Read OLS table. (1) n = resid df + #params; (2) CI $= \hat{\beta} \pm 1.96\widehat{\text{SE}}$; (3) check interaction $\Rightarrow$ main effect is conditional; (4) joint test on factors via F; (5) collinearity if huge SE + sig. F. **Read logistic table.** odds factor $= \exp(\beta\Delta)$; per-group multipliers $= \exp(\beta_x + \beta_{xz} z)$; $\hat{\eta} = \beta^\top x$ then σ; threshold vs base rate; class imbalance flags. **Read CV curve.** pick λ_{min} or λ_{1SE} (simplest); 1-SE preferred for parsimony. **Read confusion mat..** compute sens/spec; compare to base rate; threshold context. **Hierarchy/spline by hand.** write distance matrix; merge step-by-step; label y-axis. **Bias-variance derive.** state $\varepsilon \perp D$, $\mathbb{E}\varepsilon = 0$; add–subtract $\mathbb{E}\hat{f}$; cross-terms vanish; **[don't write Bias without squaring]**. **LDA derive.** $\log \pi_k f_k$; drop k-free constants (incl. $\log|\Sigma|$ if shared); $-\frac{1}{2}x^\top \Sigma^{-1} x$ drops **[because Σ shared]**; remaining linear in x.